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150

Homework — 3

Course Name: Advanced Microeconomic Theory

Course No : Econ 501

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Ans. to the Qus. NO-1

The degree of homogeneity of a function exhibits its returns to scale. If a function is homogeneous of degree 1, then there is constant returns to scale. If the degree is less than 1, then it is decreasing returns to scale, and a function with homogeneity of degree greater than 1 shows increasing returns to scale

a) $Q = K^{\frac{1}{2}} L^{\frac{1}{2}}$

$$\begin{aligned} Q &= (OK)^{\frac{1}{2}} (OL)^{\frac{1}{2}} \\ &= \underline{O^{\frac{1}{2}} K^{\frac{1}{2}}} \cdot \underline{O^{\frac{1}{2}} L^{\frac{1}{2}}} \\ &= \underline{O K^{\frac{1}{2}} L^{\frac{1}{2}}} \end{aligned}$$

It is homogeneous of degree 1.

So the function is constant returns to scale.

$$b) \quad q = 2K + L$$

$$= 2OK + OL$$

$$= \underline{O(2K + L)}$$

$$= \underline{Oq}$$

— The function q is homogeneous of degree 1.

So it is constant returns to scale.

$$c) \quad q = \frac{4KL}{K+L}$$

$$= \frac{4OKOL}{OK+OL}$$

$$= \frac{O^2 4KL}{O(K+L)}$$

$$= \underline{O \cdot \frac{4KL}{(K+L)}}$$

— The function q is homogeneous of degree 1.

So it is constant returns to scale.

$$d) \quad q = KL$$

$$= \theta K \cdot \theta L$$

$$= \theta^2 KL$$

— the function is homogeneous of degree 2.

So it shows increasing returns to scale.

$$e) \quad q = \min(K, \frac{1}{2} L)$$

$$= \min(\theta K, \frac{1}{2} \theta L)$$

$$= \theta \min(K, \frac{1}{2} L)$$

— the function is homogeneous of degree 1.

— this shows constant returns to scale.

Ans. to the Qus. NO - 2

The elasticity of substitution is given by.

$$\sigma = \frac{\text{percentage change in } K_L}{\text{percentage change in MRTS}}$$
$$= \frac{d \log K_L}{d \log MRTS}$$

a) $q = K^{\frac{1}{2}} L^{\frac{1}{2}}$

Here,

$$MP_L = \frac{\partial q}{\partial L} = \frac{1}{2} K^{\frac{1}{2}} L^{-\frac{1}{2}}$$

$$MP_K = \frac{\partial q}{\partial K} = \frac{1}{2} K^{-\frac{1}{2}} L^{\frac{1}{2}}$$

$$MRTS = \frac{MP_L}{MP_K} = \frac{\frac{1}{2} K^{\frac{1}{2}} L^{-\frac{1}{2}}}{\frac{1}{2} K^{-\frac{1}{2}} L^{\frac{1}{2}}}$$

$$= K L^{-1} = \underline{\underline{\frac{K}{L}}}$$

Thus,

$$\underline{\log \frac{K}{L}} = \log MRTS$$

$$\therefore \sigma = \frac{d \log \frac{K}{L}}{d \log MRTS} = 1$$

$$b) q = 2K + L$$

$$\text{Here, } \underline{MP_L = 1} \text{ and } \underline{MP_K = 2}$$

$$MRTS = \frac{MP_L}{MP_K} = \frac{1}{2} \quad ; \text{ a constant}$$

$$\therefore \log MRTS = \log\left(\frac{1}{2}\right) = \text{a constant}$$

$$\text{Thus, } d \log MRTS = 0$$

$$\sigma = \frac{d \log K_L}{d \log MRTS} = \frac{d \log K_L}{0} = \infty$$

$$c) q = \frac{4KL}{K+L}$$

$$MP_L = \frac{(K+L)4K - 4KL}{(K+L)^2}$$

$$= \frac{4K^2 + 4KL - 4KL}{(K+L)^2}$$

$$= \frac{4K^2}{(K+L)^2}$$

$$MP_K = \frac{(K+L)4L - 4KL}{(K+L)^2}$$

$$= \frac{4KL + 4L^2 - 4KL}{(K+L)^2}$$

$$= \frac{4L^2}{(K+L)^2}$$

$$MRTS = \frac{MP_L}{MP_K} = \frac{4K^2/(K+L)^2}{4L^2/(K+L)^2}$$

$$\Rightarrow MRTS = \frac{K^2}{L^2} = \left(\frac{K}{L}\right)^2$$

$$\Rightarrow \log MRTS = 2 \log \left(\frac{K}{L}\right)$$

$$\Rightarrow \log \left(\frac{K}{L}\right) = \frac{1}{2} \log MRTS$$

$$\therefore \sigma = \frac{d \log K/L}{d \log MRTS}$$

$$\Rightarrow \sigma = \frac{1}{2}$$

$$d) \quad q = KL$$

$$\text{Hence, } MP_L = \underline{K} \quad \text{and} \quad MP_K = \underline{L}$$

$$MRTS = \frac{MP_L}{MP_K} = \frac{K}{L}$$

$$\therefore \log\left(\frac{K}{L}\right) = \log MRTS$$

$$\text{Therefore, } \sigma = \frac{d \log\left(\frac{K}{L}\right)}{d \log MRTS} = 1.$$

$$e) \quad q = \min(K, \frac{1}{2}L)$$

$$\text{When, } K < \frac{1}{2}L, \text{ then } q = K$$

$$MP_L = 0 \quad \text{and} \quad MP_K = 1$$

$$\Rightarrow MRTS = \frac{0}{1} = 0.$$

$$\text{When, } \frac{1}{2}L < K, \text{ then } q = \frac{1}{2}L$$

$$MP_L = \frac{1}{2} \quad \text{and} \quad MP_K = 0$$

$$MRTS = \frac{MP_L}{MP_K} = \frac{\frac{1}{2}}{0} = \infty$$

the optimizing condition is

$$K = \frac{1}{2} L$$

$$\Rightarrow \frac{K}{L} = \frac{1}{2}$$

$$\Rightarrow \log \frac{K}{L} = \log \left(\frac{1}{2} \right) = \text{a constant}$$

$$\therefore d \log \frac{K}{L} = 0$$

the elasticity of substitution,

$$\sigma = \frac{d \log \frac{K}{L}}{d \log MRTS}$$

$$\Rightarrow \sigma = \frac{0}{d \log MRTS}$$

$$\therefore \sigma = 0.$$

The value of elasticity of substitution is 0.

Ans. to the Qus. NO-3

In the short run, capital is fixed at $\bar{K} = 64$.

a) $q = K^{\frac{1}{2}} L^{\frac{1}{2}}$

$$q_{SR} = 64^{\frac{1}{2}} L^{\frac{1}{2}} \\ = 8L^{\frac{1}{2}}$$

Short run cost function $C_{SR} = wL + vK$

where $w = \text{price of labor}$

$v = \text{price of capital}$

$$q = 8L^{\frac{1}{2}}$$

$$\Rightarrow L^{\frac{1}{2}} = \frac{q}{8}$$

$$\Rightarrow L = \frac{q^2}{64}$$

$$\therefore \underline{C_{SR} = w \cdot \frac{q^2}{64} + 64v}$$

$$b) q = 2K + L$$

$$q_{SR} = 2(64) + L = 128 + L$$

Then,

$$L = q - 128$$

$$C_{SR} = \underline{\omega(q - 128) + 64v}$$

$$c) q = \frac{4KL}{K+L}$$

$$\Rightarrow q_{SR} = \frac{4(64)L}{64+L}$$

$$= \frac{256L}{64+L}$$

Now,

$$q(64+L) = 256L$$

$$\Rightarrow 256L - qL = 64q$$

$$\Rightarrow L = \frac{64q}{256-q} \quad \text{for } q < 256$$

$$\therefore C_{SR} = \underline{\omega \cdot \frac{64q}{(256-q)} + 64v}$$

(6)

d) $q = KL$

$\Rightarrow q_{SR} = 64L$

Then, $L = \frac{q}{64}$

$C_{SR} = w \cdot \frac{q}{64} + 64v$

e) $q = \min(K, \frac{1}{2}L)$

$q_{SR} = \min(64, \frac{1}{2}L)$

If, $\frac{1}{2} < 64$, then, $q = \frac{1}{2} \Rightarrow L = 2q$

$C_{SR} = w(2q) + 64v$

If, $64 \leq \frac{1}{2}$ that is $L \geq 128$

then $q = 64$

$C_{SR} = 128w + 64v$ [Here, $L = 128 = 2q$]

So, $C_{SR} = w(2q) + 64v$

where

or $C_{SR} = 128w + 64v$

Ans. to the Ques. No - 4

The cost minimizing point of a firm is where the isoquant is tangent to the isocost line. The locus of these tangencies is called the firm's expansion path.

The expansion path equation of a firm is thus given

by,

$$\frac{MP_L}{MP_K} = \frac{w}{r}$$

a) $Q = K^{\frac{1}{2}} L^{\frac{1}{2}}$

$$MP_L = \frac{1}{2} K^{\frac{1}{2}} L^{-\frac{1}{2}} \quad \text{and} \quad MP_K = \frac{1}{2} K^{-\frac{1}{2}} L^{\frac{1}{2}}$$

$$\frac{MP_L}{MP_K} = \frac{K}{L} = \frac{w}{r}$$

$$\therefore K = \frac{w}{r} \cdot L \quad ; \text{ the expansion path equation.}$$

b) $q = 2K + L$

the optimizing condition is,

$$\frac{MP_L}{MP_K} = \frac{w}{v} \quad \text{or} \quad \frac{MP_L}{w} = \frac{MP_K}{v}$$

Here, $MP_L = 1$ and $MP_K = 2$

If, $\frac{MP_L}{w} = \frac{1}{w} > \frac{MP_K}{v} = \frac{2}{v}$,

the firm uses only L in production.

So the firm's expansion path becomes the axis where we are measuring L or the L-axis.

If,

$$\frac{MP_L}{w} = \frac{1}{w} < \frac{MP_K}{v} = \frac{2}{v}$$

then the firm uses only K in production.

Therefore, the firm's expansion path is given by the K-axis.

$$c) \quad a_1 = \frac{4KL}{K+L}$$

$$\text{Here, } MP_L = \frac{(K+L)4K - 4KL}{(K+L)^r} = \frac{4K^r + 4KL - 4KL}{(K+L)^r}$$

$$= \frac{4K^r}{(K+L)^r}$$

$$MP_K = \frac{(K+L)4L - 4KL}{(K+L)^r} = \frac{4KL + 4L^r - 4KL}{(K+L)^r}$$

$$= \frac{4L^r}{(K+L)^r}$$

$$\frac{MP_L}{MP_K} = \frac{4K^r / (K+L)^r}{4L^r / (K+L)^r} = \frac{K^r}{L^r}$$

$$\Rightarrow \frac{K^r}{L^r} = \frac{w}{v}$$

$$\Rightarrow K^r = \frac{w}{v} \cdot L^r$$

$$\Rightarrow K = \sqrt{\frac{w}{v}} \cdot L$$

d) $q = KL$

Here, $MP_L = K$ and $MP_K = L$

$$\frac{MP_L}{MP_K} = \frac{K}{L} = \frac{w}{v}$$

$$\Rightarrow \underline{K = \frac{w}{v} \cdot L}$$

e) $q = \min(K, \frac{1}{2}L)$

The cost minimizing condition is,

$$\underline{K = \frac{1}{2}L}$$

the inputs must be used in this proportion to produce various levels of output.

So the output expansion path is the line,

$K = \frac{1}{2}L$ ✓

~~$50L + 10K = 50L + 64 \times \frac{1}{2}L \Rightarrow L > 128$~~

~~$50L + 10K = 50(22) + 64 \times \frac{1}{2}(22) = 2200$~~

Ans. to the Qus. No-5

Here given.

$$\underline{w=2} ; \quad \underline{V=8} ; \quad \text{and} \quad \underline{q=100}$$

So, we minimize

$$\underline{C = 2L + 8K}$$

$$\text{Subject to, } \underline{q = f(K, L) = 100}$$

$$\text{a) } \underline{q = K^{\frac{1}{2}} L^{\frac{1}{2}}} \Rightarrow \underline{100 = K^{\frac{1}{2}} L^{\frac{1}{2}}}$$

$$\text{Min}_{K, L, \lambda} \quad \underline{L = 2L + 8K + \lambda [100 - K^{\frac{1}{2}} L^{\frac{1}{2}}]}$$

$$\frac{\partial L}{\partial L} = 2 - \frac{1}{2} \lambda \cdot K^{\frac{1}{2}} L^{-\frac{1}{2}} = 0 \quad \text{--- (I)}$$

$$\frac{\partial L}{\partial K} = 8 - \frac{1}{2} \lambda K^{-\frac{1}{2}} L^{\frac{1}{2}} = 0 \quad \text{--- (II)}$$

$$\frac{\partial L}{\partial \lambda} = 100 - K^{\frac{1}{2}} L^{\frac{1}{2}} = 0$$

$$\Rightarrow K^{\frac{1}{2}} L^{\frac{1}{2}} = 100 \quad \text{--- (III)}$$

from eqn ① & ⑩,

$$\frac{\frac{1}{2} \lambda K^{\frac{1}{2}} L^{-\frac{1}{2}}}{\frac{1}{2} \lambda K^{-\frac{1}{2}} L^{\frac{1}{2}}} = \frac{2}{8}$$

$$\Rightarrow \frac{K}{L} = \frac{1}{4}$$

$$\Rightarrow K = \frac{L}{4} \quad \text{--- (11)}$$

Substituting $K = \frac{L}{4}$ in ⑪,

$$\left(\frac{L}{4}\right)^{\frac{1}{2}} L^{\frac{1}{2}} = 100$$

$$\Rightarrow \frac{L}{2} = 100$$

$$\Rightarrow L^* = 200$$

Thus,

$$K = \frac{L}{4}$$

$$= \frac{200}{4}$$

$$\therefore K^* = 50$$

The cost minimizing input mix is,

$$L^* = 200 \text{ and } K^* = 50.$$

b) $q = 2K + L = 100$

Here, $MP_L = 1$ and $MP_K = 2$

Given, $w = 2$ and $v = 8$

Thus,

$$\frac{MP_L}{w} = \frac{1}{2} > \frac{MP_K}{v} = \frac{2}{8} = \frac{1}{4}$$

So the firm employs only L in production.

Implying $K^* = 0$

the output function,

$$q = 2(0) + L$$

$$\Rightarrow L^* = 100.$$

The cost minimizing input mix is,

$K^* = 0$ and $L^* = 100$.

$$c) \quad q = \frac{4KL}{K+L} = 100$$

$$\text{Min}_{K, L, \lambda} \quad \mathcal{L} = 2L + 2K + \lambda \left[100 - \frac{4KL}{K+L} \right]$$

First order condition requires,

$$\frac{\partial \mathcal{L}}{\partial L} = 2 - \lambda \cdot \frac{(K+L)4K - 4KL}{(K+L)^2} = 0 \quad \text{--- (I)}$$

$$\frac{\partial \mathcal{L}}{\partial K} = 2 - \lambda \cdot \frac{(K+L)4L - 4KL}{(K+L)^2} = 0 \quad \text{--- (II)}$$

$$\frac{\partial \mathcal{L}}{\partial \lambda} = 100 - \frac{4KL}{K+L} = 0$$

$$\Rightarrow \frac{4KL}{K+L} = 100 \quad \text{--- (III)}$$

from (I) & (II),

$$\frac{\lambda \cdot 4K^2 / (K+L)^2}{\lambda \cdot 4L^2 / (K+L)^2} = \frac{2}{2}$$

$$\Rightarrow \frac{K^2}{L^2} = 1$$

$$\Rightarrow K^2 = L^2$$

$$\Rightarrow K = L$$

Substituting $k = \frac{L}{2}$ into (11) $\Rightarrow \frac{4 \cdot \frac{L}{2} \cdot L}{\frac{L}{2} + L} = 100$

$$\frac{4 \cdot \frac{L}{2} \cdot L}{\frac{L}{2} + L} = 100$$

$$\Rightarrow \frac{2L^2}{\frac{3}{2}L} = 100$$

$$\Rightarrow 2L = 100 \times \frac{3}{2}$$

$$\Rightarrow L = \frac{100}{2} \times \frac{3}{2}$$

$$\Rightarrow \underline{L^* = 75}$$

Therefore, $k = \frac{L}{2}$

$$\Rightarrow k^* = \frac{75}{2} = 37.5$$

The cost minimizing input mix is,

$$\underline{L^* = 75} \text{ and } \underline{k^* = 37.5}$$

(11)

$$d) q = KL \Rightarrow 100 = KL$$

$$\text{Min}_{K, L, \lambda} \quad \mathcal{L} = 2L + 8K + \lambda [100 - KL]$$

first order condition requires,

$$\frac{\partial \mathcal{L}}{\partial L} = 2 - \lambda K = 0 \Rightarrow 2 = \lambda K \quad \text{--- (i)}$$

$$\frac{\partial \mathcal{L}}{\partial K} = 8 - \lambda L = 0 \Rightarrow 8 = \lambda L \quad \text{--- (ii)}$$

$$\frac{\partial \mathcal{L}}{\partial \lambda} = 100 - KL = 0$$

$$\Rightarrow KL = 100 \quad \text{--- (iii)}$$

from (i) and (ii),

$$\frac{\lambda K}{\lambda L} = \frac{2}{8}$$

$$\Rightarrow \frac{K}{L} = \frac{1}{4}$$

$$\Rightarrow K = \frac{L}{4}$$

Substituting $K = \frac{L}{4}$ into (iii).

$$\frac{L}{4} \cdot L = 100$$

$$\Rightarrow \frac{L^2}{4} = 100$$

$$\Rightarrow L^2 = 400$$

$$\Rightarrow \underline{L^* = 20}$$

— Thus, $K = \frac{L}{4}$

$$\Rightarrow K = \frac{20}{4}$$

$$\Rightarrow \underline{K^* = 5.}$$

(12)

$$e) \quad \underline{q = \min(K, \frac{1}{2}L) = 100}$$

If, $K < \frac{1}{2}L$, then $q = K$, that is $K = 100$.

If, $\frac{1}{2} < K$, then, $q = \frac{1}{2}L \Rightarrow L = 2q = 200$.

In the cost minimizing level,

$$K = \frac{1}{2}L = q$$

that is,

$$K = q$$

$$\Rightarrow K^* = 100$$

$$\text{and, } \frac{1}{2}L = q$$

$$\Rightarrow L = 2q$$

$$\Rightarrow L^* = 200$$

The cost minimizing input mix is therefore,

$$L^* = 200 \quad \text{and} \quad K^* = 100.$$

Ans. to the Qus. NO-6

The long-run cost function is given by

$$\underline{C = wL + vK}$$

where,

$w = \text{price of labor}$

$v = \text{price of capital.}$

a) $q = K^{\frac{1}{2}} L^{\frac{1}{2}}$

$$\text{Min}_{K, L, \lambda} \quad L = wL + vK + \lambda [q - K^{\frac{1}{2}} L^{\frac{1}{2}}]$$

FOC requires,

$$\frac{\partial L}{\partial L} = w - \frac{1}{2} \lambda K^{\frac{1}{2}} L^{-\frac{1}{2}} = 0 \quad \text{--- (i)}$$

$$\frac{\partial L}{\partial K} = v - \frac{1}{2} \lambda K^{-\frac{1}{2}} L^{\frac{1}{2}} = 0 \quad \text{--- (ii)}$$

$$\frac{\partial L}{\partial \lambda} = q - K^{\frac{1}{2}} L^{\frac{1}{2}} = 0$$

$$\Rightarrow q = K^{\frac{1}{2}} L^{\frac{1}{2}} \quad \text{--- (iii)}$$

from ① and ②

$$\frac{w}{v} = \frac{\frac{1}{2} \lambda K^{\frac{1}{2}} L^{-\frac{1}{2}}}{\frac{1}{2} \lambda K^{-\frac{1}{2}} L^{\frac{1}{2}}}$$

$$\Rightarrow \frac{K}{L} = \frac{w}{v}$$

$$\Rightarrow K = \frac{w}{v} \cdot L \quad \text{--- ③}$$

Substituting eqn ③ into ②,

$$\left(\frac{w}{v} \cdot L\right)^{\frac{1}{2}} L^{\frac{1}{2}} = q$$

$$\Rightarrow L \cdot \left(\frac{w}{v}\right)^{\frac{1}{2}} = q$$

$$\Rightarrow L^* = \frac{q \cdot \sqrt{\frac{v}{w}}}{1}$$

$$K^* = \frac{w}{v} \cdot q \sqrt{\frac{v}{w}}$$

$$= q \sqrt{\frac{w}{v}}$$

$$C = w \cdot q \sqrt{\frac{v}{w}} + v \cdot q \sqrt{\frac{w}{v}}$$

$$= q \sqrt{wv} + q \sqrt{wv}$$

$$= 2q \sqrt{w \cdot v}$$

$$b) \quad \underline{q = 2K + L}$$

Cost minimizing condition is, $\frac{MP_L}{MP_K} = \frac{w}{v}$
or, $\frac{MP_L}{w} = \frac{MP_K}{v}$.

Here, $\underline{MP_L = 1}$ and $\underline{MP_K = 2}$

If, $\underline{\frac{1}{w} > \frac{2}{v}}$, the firm will employ only L .

Then, $\underline{K^* = 0}$.

output function, $\underline{q = 2(0) + L}$

$$\Rightarrow \underline{L^* = q}$$

the cost function, $\underline{C = wL = q \cdot w}$

If, $\underline{\frac{2}{v} > \frac{1}{w}}$, the firm will employ only K .

Then, $\underline{L^* = 0}$

the output function, $\underline{q = 2K + 0 \Rightarrow q = 2K}$

$$\Rightarrow \underline{K^* = \frac{q}{2}}$$

the cost function,

$$\underline{C = vK = \frac{q}{2} \cdot v}$$

$$c) q = \frac{4KL}{K+L}$$

$$\text{Min.}_{K, L, \lambda} \quad \mathcal{L} = wL + vK + \lambda \left[q - \frac{4KL}{K+L} \right]$$

FOC requires,

$$\frac{\partial \mathcal{L}}{\partial L} = w - \lambda \cdot \frac{(K+L) \cdot 4K - 4KL}{(K+L)^2} = 0 \quad \text{--- (I)}$$

$$\frac{\partial \mathcal{L}}{\partial K} = v - \lambda \cdot \frac{(K+L) \cdot 4L - 4KL}{(K+L)^2} = 0 \quad \text{--- (II)}$$

$$\frac{\partial \mathcal{L}}{\partial \lambda} = q - \frac{4KL}{K+L} = 0$$

$$\Rightarrow \frac{4KL}{K+L} = q \quad \text{--- (III)}$$

from (I) and (II),

$$\frac{w}{v} = \frac{\lambda \cdot 4K^2 / (K+L)^2}{\lambda \cdot 4L^2 / (K+L)^2}$$

$$\Rightarrow \frac{K^2}{L^2} = \frac{w}{v}$$

$$\Rightarrow K^2 = \frac{w}{v} \cdot L^2$$

$$\Rightarrow K = \sqrt{\frac{w}{v}} \cdot L \quad \text{--- (IV)}$$

Substituting (iv) into (iii),

$$\frac{4 \cdot \sqrt{\frac{w}{v}} L \cdot L}{\sqrt{\frac{w}{v}} L + L} = q$$

$$\Rightarrow \frac{4L^2 \sqrt{\frac{w}{v}}}{L(\sqrt{\frac{w}{v}} + 1)} = q$$

$$\Rightarrow L^* = \frac{q(\sqrt{\frac{w}{v}} + 1)}{4\sqrt{\frac{w}{v}}} \checkmark$$

Now,

$$K^* = \frac{\cancel{4} \sqrt{\frac{w}{v}} \cdot \frac{q(\sqrt{\frac{w}{v}} + 1)}{4 \cdot \cancel{4} \sqrt{\frac{w}{v}}}}{= \frac{q}{4} (\sqrt{\frac{w}{v}} + 1)} \checkmark$$

the cost function,

$$C = \underline{w \cdot \frac{q(\sqrt{\frac{w}{v}} + 1)}{4\sqrt{\frac{w}{v}}}} + v \cdot \underline{\frac{q}{4} (\sqrt{\frac{w}{v}} + 1)}$$

$$C_{LR} = \frac{q}{4} (\sqrt{w} + \sqrt{v})$$

$$\begin{aligned} K &= L \cdot \sqrt{\frac{w}{v}} \\ L &= K \cdot \sqrt{\frac{v}{w}} \\ q &= \frac{4 \cdot K \cdot K \sqrt{\frac{v}{w}}}{K + K \sqrt{\frac{v}{w}}} \\ K^* &= \frac{q(1 + \sqrt{\frac{v}{w}})}{4 \cdot \sqrt{\frac{v}{w}}} \end{aligned}$$

d) $q = KL$

Min $\mathcal{L} = wL + vK + \lambda [q - KL]$
 K, L, λ

FOC requires,

$$\frac{\partial \mathcal{L}}{\partial L} = w - \lambda K = 0 \Rightarrow w = \lambda K \text{ --- (i)}$$

$$\frac{\partial \mathcal{L}}{\partial K} = v - \lambda L = 0 \Rightarrow v = \lambda L \text{ --- (ii)}$$

$$\frac{\partial \mathcal{L}}{\partial \lambda} = q - KL = 0 \Rightarrow KL = q \text{ --- (iii)}$$

from eqn (i) and (ii),

$$\frac{\lambda K}{\lambda L} = \frac{w}{v}$$

$$\Rightarrow \frac{K}{L} = \frac{w}{v}$$

$$\Rightarrow K = \frac{w}{v} \cdot L \text{ --- (iv)}$$

Substituting (iv) into (iii) we get,

$$\frac{w}{v} \cdot L \cdot L = q$$

$$\Rightarrow L^2 = q \cdot \frac{v}{w}$$

$$\Rightarrow L^* = \sqrt{q \cdot \frac{v}{w}}$$

thus,

$$K = \frac{w}{v} \cdot L$$

$$\Rightarrow K = \frac{w}{v} \cdot \sqrt{q \cdot \frac{v}{w}}$$

$$\Rightarrow K^* = \sqrt{q \cdot \frac{w}{v}}$$

the cost function,

$$C = w \cdot \sqrt{q \cdot \frac{v}{w}} + v \cdot \sqrt{q \cdot \frac{w}{v}}$$

$$= \sqrt{q \cdot w \cdot v} + \sqrt{q \cdot v \cdot w}$$

$$= 2\sqrt{q \cdot w \cdot v}$$

$$e) \quad q = \min(K, \frac{1}{2}L)$$

In the long run, no input should be wasted. This

implies,

$$K = \frac{1}{2}L = q$$

$$\text{— thus, } K^* = q$$

$$\text{and, } \frac{1}{2}L = q$$

$$\Rightarrow L^* = 2q$$

the cost function,

$$c = w \cdot 2q + v \cdot q$$

$$= q(2w + v)$$